

»» Chapter XVII ««

SOME QUESTIONS CONCERNING UNIVALENT ANALYTIC FUNCTIONS

PREFATORY REMARKS

From the very rich field of univalent analytic functions this chapter has drawn but three topics. Our choice was motivated by two considerations: one, to select material that ought to be in the repertory of every doctoral candidate in analysis, the other, to select material illustrating and applying the principles learned thus far in the book without appeal to outside resources. We have chosen the Riemann mapping theorem, Carathéodory's theorem on variable regions, and the application of the Schwarz reflexion principle in the study of boundary behavior of univalent analytic functions under special conditions concerning boundary behavior. The subject of univalent analytic functions is vast and has many aspects. For one interested in the boundary behavior of functions analytic on $\Delta(0; 1)$ which are univalent and of their inverses the theory of the Lebesgue integral and a refined study of the topology of the plane are essential. The two-volume report of Gattegno and Ostrowski [41] is an excellent source for the study of boundary questions. For coefficient problems the monograph of Jenkins [62] is to be recommended. For an over-all exposition with strong emphasis on the geometric side the treatise of Golusin [43] is valuable.

1. THE RIEMANN MAPPING THEOREM (PLANE CASE)

This theorem may be stated as follows.

Theorem 1.1: *Let A be a holomorphically simply connected region which is a proper part of $\mathbb{C}$. Then there exists a function analytic on A , univalent, and having image $\Delta(0; 1)$.*

PROOF: The following proof is due to Fejér and F. Riesz. Let $a \in A$. Let Φ denote the set of functions f analytic on A , univalent, having image in $\Delta(0; 1)$ and satisfying $f(a) = 0$. We shall show that Φ is not empty, that there exists $g \in \Phi$ such that $|g'(a)| \geq |f'(a)|$, $f \in \Phi$, and that g having this property is a univalent map of A onto $\Delta(0; 1)$.

Φ is not empty. To see this we proceed as follows. Let $b \in \mathbb{C} - A$. Let l be an analytic logarithm of $z - b$ with domain A . Now l is not only univalent but also has the property that $l(w) \neq l(z) + 2\pi i$, $z, w \in A$. Hence the translate of $l(A)$ through $2\pi i$ is contained in $\mathbb{C} - l(A)$. Let S denote a Möbius transformation taking $l(a)$ into 0 and $l(a) + 2\pi i$ into ∞ . Then $cS \circ l \in \Phi$ for c a sufficiently small nonzero complex number.

There exists $g \in \Phi$ such that $|g'(a)| \geq |f'(a)|$, $f \in \Phi$. Since Φ is a normal family, there exists $(g_n): \mathbb{N} \rightarrow \Phi$ such that

$$(1) \quad \lim_{n \rightarrow \infty} |g_n'(a)| = \sup\{|f'(a)|: f \in \Phi\},$$

(2) (g_n) tends to a function g uniformly in A . It is to be noted that g so obtained is analytic on A and satisfies:

$$g(a) = 0; \quad |g'(a)| \geq |f'(a)| > 0, \quad f \in \Phi; \quad |g(z)| < 1, \quad z \in A.$$

To see that the third condition is fulfilled, we observe that at all events $|g(z)| \leq 1$, $z \in A$, and $g(a) = 0$ so that by the maximum principle for analytic functions the strong inequality holds. To see that $g \in \Phi$, we note that if a sequence of functions analytic on a region A and univalent tends uniformly in A to a nonconstant function, then this function is univalent (cf. Ex. 4.38, Ch. VI).

The image of g is $\Delta(0; 1)$. We shall introduce an auxiliary function to establish this. Given $b \in \Delta(0; 1)$ we use L_b (cf. §§3, 5, and 7, Ch. VIII), to denote the Möbius transformation given by $L_b(z) = (b - z)/(1 - \bar{b}z)$, $z \in \mathbb{C}$, $L_b(\infty) = 1/\bar{b}$. We note that the inverse of L_b is L_b itself. Suppose that $b \neq 0$ and that $c^2 = b$. We introduce $B = L_b \circ (L_c)^2$. Since B has constant valence 2, $B(0) = 0$, and $|B(z)| < 1$ for $|z| < 1$, we see by the Schwarz lemma that $|B'(0)| < 1$. Suppose now that $b \in \Delta(0; 1)$ is not in the image of g . Of course, $b \neq 0$. We let s denote the analytic square root of $L_b \circ g$ satisfying $s(a) = c$ and

thereupon define h as $L_c \circ s$. The steps leading to h insure that $h \in \Phi$. Indeed, $L_b \circ g$ is univalent and analytic on A and its image is in $\Delta(0; 1) - \{0\}$; s enjoys these properties also and hence since $s(a) = c$, $h \in \Phi$. Now $g = B \circ h$. Hence $g'(a) = B'(0) h'(a)$ and we conclude $|g'(a)| < |h'(a)|$. Contradiction. Hence the image of g is $\Delta(0; 1)$. The theorem is thereby established.

EXERCISES

1.1. Show that the totality of f analytic on A , univalent, and having image $\Delta(0; 1)$ consists of the functions of the form $L_b \circ (\eta g)$ where $\eta \in C(0; 1)$, $b \in \Delta(0; 1)$. Show that there is a unique such f satisfying $f(a) = b$, $\text{sg}[f'(a)] = \eta$, where b and η are given and fulfill the above conditions.

1.2. Let Ψ denote the set of functions analytic on A and mapping A into $\Delta(0; 1)$. Show that $|g'(a)| \geq |f'(a)|$, $f \in \Psi$, and that equality occurs precisely for f of the form ηg , $\eta \in C(0; 1)$. Let Ψ_0 denote the subset of Ψ consisting of $f \in \Psi$ satisfying $f(a) = 0$. Let $b (\neq a) \in A$. Show that $|g(b)| \geq |f(b)|$, $f \in \Psi_0$, and examine the case of equality. Here reference is made to g of Theorem 1.1.

1.3. Show with the aid of the Riemann mapping theorem that a holomorphically simply connected region A in $\mathbb{C}$ is homotopically simply connected and has the property that $\hat{\mathbb{C}} - A$ is connected. Hint: For the latter property show that

$$\hat{\mathbb{C}} - g[\Delta(0; r)],$$

$0 < r < 1$, is connected, when g is a univalent function analytic on $\Delta(0; 1)$ and use the standard theorem valid for a Hausdorff space which states that $\bigcup_{n \in \mathbb{N}} K_n$ is connected when $(K_n)_{n \in \mathbb{N}}$ is a sequence of compact connected subsets of the space satisfying $K_{n+1} \subset K_n$, $n \in \mathbb{N}$.

1.4. Show that the zeros of B (of the text) are 0 and $d = L_c(-c)$ and that $B = \eta IL_d$ for some $\eta \in C(0; 1)$ where I is the identity map of $\hat{\mathbb{C}}$. Conclude that

$$|B'(0)| = \frac{2|b|^{1/2}}{1 + |b|}.$$

Show with the aid of this formula for $|B'(0)|$ that the sequence (φ_n) of univalent functions analytic on A defined below tends uniformly in A to a function with image $\Delta(0; 1)$. Let $\varphi \in \Phi$ have image not equal to $\Delta(0; 1)$. The sequence $(\varphi_n): \mathbb{N} \rightarrow \Phi$ is defined recursively as follows: $\varphi_0 = \varphi$, $\varphi_{n+1} = \psi_n \circ \varphi_n$, $n \in \mathbb{N}$. Here ψ_n is a univalent function analytic on $\varphi_n(A)$ satisfying $\psi_n(0) = 0$, $\psi'_n(0) > 0$, $B_n \{\text{sg}[B'_n(0)] \psi'_n(z)\} = z$, $z \in \varphi_n(A)$, where B_n is just B of the text with b a point of $\mathbb{C} - \varphi_n(A)$ nearest 0 (cf. the construction of h). The possibility of constructing such a sequence is to be verified. Hint: Show that $|\varphi_{n+1}(z)| \geq |\varphi_n(z)|$, $z \in A$, and show that the sequence (r_n) is increasing and has limit 1, r_n being $\min\{|z|: z \in \mathbb{C} - \varphi_n(A)\}$.

1.5. Let f be analytic on $\Delta(0; r)$, $r > 1$, and suppose further that $f|_{\overline{\Delta(0; 1)}}$ is univalent and that $f'(z) \neq 0$, $|z| = 1$. Show that $f|_{\Delta(0; s)}$ is univalent for some s , $1 < s < r$, and that for such s a function analytic on $f[\Delta(0; s)]$ may be approximated uniformly on $f[\Delta(0; 1)]$ by a polynomial. Conclude that if h is continuous on $f[\Delta(0; 1)]$ and is analytic in $f[\Delta(0; 1)]$, then h may be approximated uniformly on $f[\Delta(0; 1)]$ by a

polynomial. Hint: Consider $h \circ [f | \overline{\Delta(0; 1)}]$ and approximate it first by a polynomial P and then consider P composed with the inverse of $f | \overline{\Delta(0; 1)}$.

Far-reaching generalizations of approximation theorems of this kind are known (cf. [83], [125]).

2. CARATHÉODORY'S THEOREM ON VARIABLE REGIONS

Let $(A_n)_{n \in \mathbb{N}}$ be a sequence of simply connected regions of $\mathbb{C}$ satisfying $0 \in A_n, n \in \mathbb{N}$.† Carathéodory defined the *kernel* of (A_n) , which we shall denote $k[(A_n)]$, as follows. If no $\Delta(0; r)$ lies in all A_n , then by $k[(A_n)]$ is meant simply $\{0\}$. Otherwise $k[(A_n)]$ is defined as $\bigcup_{n \in \mathbb{N}} B_n$ where B_n is the component of $\text{int}(\bigcap_{k \geq n} A_k)$ containing 0. His object in introducing this concept was to treat the following problem [22, pp. 362–405]:

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of univalent functions analytic on $\Delta(0; 1)$ and normalized by the conditions $f_n(0) = 0, f'_n(0) > 0, n \in \mathbb{N}$. We let A_n denote the image of f_n . We seek a necessary and sufficient condition in terms of (A_n) that (f_n) tend to some function f uniformly in $\Delta(0; 1)$.

Carathéodory solved this problem (loc. cit.) by the following theorem.

Theorem 2.1: *If (f_n) tends to f uniformly in $\Delta(0; 1)$, then the image of f is $k[(A_n)]$.*

A necessary and sufficient condition that (f_n) tend to some function f uniformly in $\Delta(0; 1)$ is that every subsequence of (A_n) have the same kernel and it be a proper subset of $\mathbb{C}$.

PROOF: Suppose that (f_n) tends uniformly to f in $\Delta(0; 1)$. Then either f is the constant zero or f is univalent. In either case let A denote the image of f . We show that $A = k[(A_n)]$, thereby establishing the first assertion of the theorem and rendering the necessity part of the second assertion immediate.

We first verify that $A \subset k[(A_n)]$. Suppose that $a \in A$. The case, $a = 0$, is trivial and we put it aside. Let b satisfy $f(b) = a (\neq 0)$ and let $|b| < \rho < r < 1$. We have

$$f[\Delta(0; \rho)] \subset f_n[\Delta(0; r)],$$

n large, as we see by Rouché's theorem applied to

$$f_n(z) - w = [f(z) - w] + [f_n(z) - f(z)],$$

† We do not need to qualify "simply connected" since we now know that the various kinds of simple connectivity used in this book for regions of $\mathbb{C}$ are equivalent, that is, holomorphic, homotopic, and simple connectivity defined by the stipulation that the complement with respect to $\hat{\mathbb{C}}$ is connected.

with domain $\overline{\Delta(0; r)}$, $w \in f[\Delta(0; \rho)]$, n large. Hence $a \in k[(A_n)]$ and we conclude that $A \subset k[(A_n)]$.

We next show that $k[(A_n)] \subset A$. We put aside the trivial case $k[(A_n)] = \{0\}$. Let g_n denote the inverse of f_n . Let $\overline{\Delta(0; r)} \subset A_n$ for all $n \in \mathbb{N}$, where $0 < r < +\infty$. By the minimum principle for the modulus of an analytic function applied to $f_n(z)/z$ and the univalence of f_n we have $|f_n(z)| \geq r|z|$ for $|z| < 1$.† Hence $f_n[\Delta(0; 1/2)] \supset \Delta(0; r/2)$. Using the integral formula representing $g_n|_{\Delta(0; r/2)}$ in terms of $f_n|_{C(0; 1/2)}$ and $f_n'|_{C(0; 1/2)}$, we see that $(g_n|_{\Delta(0; r/2)})$ is uniformly Cauchy in $\Delta(0; r/2)$. Since the domain of g_n contains B_m when $n \geq m$, by the theorem of Stieltjes-Vitali (Ex. 1.14, Ch. XIII), $(g_{n+m}|_{B_m})$ tends uniformly in B_m to a function $g^{[m]}$, say. Let $g = \bigcup_{m \in \mathbb{N}} g^{[m]}$. We have: g is analytic on $k[(A_n)]$, $g(0) = 0$, the image of g is contained in $\Delta(0; 1)$. From $f_n[g_n(z)] = z$, $z \in A_n$, we conclude that $f[g(z)] = z$, $z \in k[(A_n)]$. Hence $k[(A_n)] \subset A$.

Putting the results of the two preceding paragraphs together we see that $A = k[(A_n)]$. Now $A \neq \mathbb{C}$, otherwise Liouville's theorem applied to the inverse of f would furnish a contradiction.

To complete the proof of the theorem, there remains to show that the sufficiency part of the second assertion holds. We introduce μ_n , the distance from $\{0\}$ to $\mathbb{C} - A_n$. The sequence (μ_n) is bounded, otherwise for some subsequence $(A_{m(n)})$ of (A_n) we would have $k[(A_{m(n)})] = \mathbb{C}$. We introduce Φ the set of univalent functions f , analytic on $\Delta(0; 1)$ and satisfying:

$$f(0) = 0, \quad \min\{|w| : w \neq f(z), |z| < 1\} = 1,$$

and note that by (1.8) of Ch. XIII there exists $M :]0, 1[\rightarrow \mathbb{P}$ such that

$$|z| \leq |f(z)| \leq |z| M(r), \quad (2.1)$$

$|z| \leq r < 1$, $f \in \Phi$. Applying (2.1) to $\mu_n^{-1}f_n$ we see that $\{f_n : n \in \mathbb{N}\}$ is a normal family. The proof will follow once we show that (f_n) tends to a limit pointwise on $\Delta(0; 1)$. If this were not the case, there would exist two subsequences of (f_n) tending uniformly in $\Delta(0; 1)$ to distinct functions, say g and h . Of these one is not the constant 0. By the first part of the theorem and the sufficiency hypothesis, the images of g and h are both equal to $k[(A_n)]$. Hence neither is the constant 0. Further, $h^{-1} \circ g$ is a conformal automorphism of $\Delta(0; 1)$ taking 0 into itself and having a positive derivative at 0. Hence $h^{-1} \circ g$ is the identity map on $\Delta(0; 1)$ and we conclude that $h = g$. Contradiction. The proof is complete.

† We may use Theorem 4.1, Ch. VI in place of the minimum principle.

EXERCISE

2.1. Continuing with the setting of Theorem 1.1 let g be analytic on $\Delta(0; 1)$ and in addition to satisfying $g(0) = 0$ either be the constant 0 or be univalent and satisfy $g'(0) > 0$. Then (f_n) tends uniformly to g in $\Delta(0; 1)$ if the image of g is the kernel of every subsequence of (A_n) .

3. APPLICATION OF THE SCHWARZ REFLEXION PRINCIPLE

The notion of a *regular, Jordan, analytic arc* is introduced as follows. A path γ with domain $[a, b]$, $-\infty < a < b < +\infty$, is termed an *analytic arc* provided that $\gamma = \varphi| [a, b]$ where φ is analytic on an open set of $\mathbb{C}$ containing $[a, b]$. The term *Jordan* is adjoined when γ is univalent, the term *regular* when $\gamma'(t) \neq 0$, $a \leq t \leq b$, one-sided derivatives being understood at the endpoints. Suppose now that f is analytic on $\{\operatorname{Im} z > 0\}$ and that for each $c \in O$ where O is an open set of $\mathbb{R}$ there exist $r_c > 0$ and a regular Jordan analytic arc γ_c with the property that, for each real x , $|x - c| < r_c$, $\lim_x d(z) = 0$, where $d(z)$ is the distance from $\{f(z)\}$ to γ_c . We shall show

Theorem 3.1: *Under the stated assumptions f is the restriction to $\{\operatorname{Im} z > 0\}$ of a function analytic at each point of $O \cup \{\operatorname{Im} z > 0\}$.*

PROOF: Given c , we suppose, as we may, that γ_c is the restriction to its domain $[a, b]$ of a univalent function φ analytic on a region containing $[a, b]$, and that the domain of φ is symmetric with respect to the real axis. We take ρ , $0 < \rho < r_c$, so that $S = \Delta(c; \rho) \cap \{\operatorname{Im} z > 0\}$ is mapped by f into the image of φ . The Schwarz reflexion principle applies to $\varphi^{-1} \circ (f|S)$ and shows that $\varphi^{-1} \circ (f|S)$ is the restriction to S of a function ψ analytic on $\Delta(c; \rho)$ whose image lies in the domain of φ . Hence $f|S = (\varphi \circ \psi)|S$ and the assertion of the theorem follows.

Another application of the Schwarz reflexion principle. Suppose that $\Gamma: C(0, 1) \rightarrow \mathbb{C}$ is univalent and is the restriction to $C(0; 1)$ of a function analytic at each point of $C(0; 1)$ and that $\Gamma'(z) \neq 0$, $|z| = 1$. By a *closed, regular, analytic, Jordan curve* we understand the image of such a Γ and we shall, as in similar questions, denote image and map by the same symbol. We suppose, as we may, that Γ is the restriction to $C(0; 1)$ of a univalent function φ analytic on an annulus $\{r < |z| < 1/r\}$, $0 < r < 1$. Suppose that f is a univalent function analytic on $\Delta(0; 1)$, the frontier of whose image is Γ . We shall prove the following theorem concerning the boundary behavior of f .

Theorem 3.2: *Under the stated assumptions on Γ and f there exists a univalent function F analytic on some $\Delta(0; s)$, $s > 1$, such that $f = F| \Delta(0; 1)$.*

PROOF: By Theorem 4.1, Ch. VI, we see that Γ is the set of cluster values of f . Hence since f is analytic, the component of $\hat{\mathbb{C}} - \Gamma$ containing ∞ has no points in common with the image of f , which is consequently bounded. Further let ρ , $0 < \rho < 1$, be so near 1 that $f(\{\rho < |z| < 1\})$ is contained in the image of φ . Since it is connected, $f(\{\rho < |z| < 1\})$ lies either in $\varphi(\{r < |z| < 1\})$ or in $\varphi(\{1 < |z| < 1/r\})$. Now let ψ denote the inverse of φ and consider the harmonic function

$$\log |\psi[f(e^{iz})]|$$

with domain $\{0 < \text{Im } z < \log(1/\rho)\}$. By the Schwarz reflexion principle and the periodicity of the exhibited function we see that $\log |\psi[f(z)]| = u(z)$, $\rho < |z| < 1$, where u is harmonic on $\{\rho < |z| < 1/\rho\}$ and $u(z) = 0$, $|z| = 1$. Since the logarithmic derivative of the function $\psi[f(z)]$ with domain $\{\rho < |z| < 1\}$ is the restriction to its domain of $2u_z$, we conclude that $2u_z$ is the logarithmic derivative of a function analytic on $\{\rho < |z| < 1/\rho\}$ (cf. Ex. 1.11, Ch. VI). Hence there exists a function g analytic on $\{\rho < |z| < 1/\rho\}$ satisfying $g(z) = \psi[f(z)]$, $\rho < |z| < 1$. Further $|\log |g(z)|| < \log(1/r)$, $\rho < |z| < 1/\rho$. Consequently $f \cup (\varphi \circ g)$ is analytic on $\Delta(0; 1/\rho)$. The assertion of the theorem will follow once we show that $g|C(0; 1)$ is univalent and $g'(z) \neq 0$, $|z| = 1$.

Suppose that g has multiplicity $n > 1$ at $a \in C(0; 1)$. From

$$\lim_{t \rightarrow 0} t^{-n} [|g(a + te^{i\theta})|^2 - |g(a)|^2] = \text{Re}(Ae^{ni\theta}),$$

where $A \in \mathbb{C} - \{0\}$, we conclude that g takes values of modulus greater than one and of modulus less than one in $\{\rho < |z| < 1\}$ at points arbitrarily close to a .† This is not possible by the property of $f(\{\rho < |z| < 1\})$ noted in the preceding paragraph. Hence $g'(z) \neq 0$, $|z| = 1$.

Suppose now that $g(a) = g(b)$, where $|a| = |b| = 1$. Suppose that $\lim_{n \rightarrow \infty} a_n = a$ where $\rho < |a_n| < 1$. We suppose, as we may, that (a_n) is so chosen that $g(a_n) = g(b_n)$, $n \in \mathbb{N}$, where $\lim_{n \rightarrow \infty} b_n = b$. Since either

$$g(\{\rho < |z| < 1\}) \subset \{r < |z| < 1\},$$

in which case

$$g(\{1 \leq |z| < 1/\rho\}) \subset \{1 \leq |z| < 1/r\},$$

or else

$$g(\{\rho < |z| < 1\}) \subset \{1 < |z| < 1/r\},$$

† Here we consider t satisfying $0 < t < 1 - \rho$.

in which case

$$g(\{1 \leq |z| < 1/\rho\}) \subset \{r < |z| \leq 1\},$$

we see that $\rho < |b_n| < 1$. By the univalence of f we see that $a_n = b_n$, $n \in \mathbf{N}$. Consequently $a = b$. The proof is complete.

EXERCISE

3.1. Apply Theorem 3.2 to show that the factor $\frac{1}{2}$ occurring in the theorem of Fejér and F. Riesz (Ex. 3.30, Ch. V) cannot be improved. For this purpose take Γ as the ellipse

$$\{(z + z^{-1})/2 : |z| = r\}, \quad 1 < r < +\infty.$$

Let g denote a function of the form guaranteed by Theorem 3.2 whose restriction to $\Delta(0; 1)$ is the univalent function h analytic on $\Delta(0; 1)$ which has as its image the bounded region A satisfying $\text{fr } A = \Gamma$, and which is such that $h(0) = 0$, $h'(0) > 0$. Let f denote an analytic square root of g' . Show that the ratio of the two integrals of the theorem of Fejér and F. Riesz is arbitrarily close to $\frac{1}{2}$ when r is sufficiently near one.

4. SCHWARZ-CHRISTOFFEL FUNCTIONS

We shall be concerned with characterizing the continuous maps F of $\{\text{Im } z \geq 0\} \cup \{\infty\}$ into $\mathbf{C}$ which satisfy the condition that there exists an increasing sequence $(a_k)_{1 \leq k \leq n}$ of real numbers such that (1) F is the restriction to its domain of a function analytic at each point of

$$\{\text{Im } z \geq 0\} - \{a_k : 1 \leq k \leq n\}$$

and having a nonzero derivative at each such point, (2) the images with respect to F of $[a_k, a_{k+1}]$, $k = 1, \dots, n-1$, as well as of $\{x \leq a_1\} \cup \{\infty\}$ and of $\{x \geq a_n\} \cup \{\infty\}$ are segments in $\mathbf{C}$. We term such F *Schwarz-Christoffel functions*. To be sure, one customarily considers univalent F . However, the standard necessary condition that arises in the univalent theory is not sufficient in that theory, but it is a characterizing condition for the broader class considered here. The treatment of the univalent case is subject to inevitable complications.

Let f denote the restriction to $\{\text{Im } z > 0\}$ of an admitted F . On considering an interval I in $\mathbf{R} - \{a_k : 1 \leq k \leq n\}$ we see that there exist $\eta, \alpha \in \mathbf{C}$, $\eta \neq 0$, such that $\text{Im}[\eta F(x) + \alpha] = 0$, $x \in I$. Hence $F''(x)/F'(x)$ is real. It follows that f''/f' is the restriction to $\{\text{Im } z > 0\}$ of a function ϕ analytic on $\mathbf{C} - \{a_k : 1 \leq k \leq n\}$.

Let us now consider the behavior of φ near ∞ and the a_k . To that end, we first show that v_f , the valence of f , is *bounded*. Otherwise $\limsup_w v_f = +\infty$ for some $w \in \mathbb{C}$ and we conclude that there exists a sequence $(w_n): \mathbb{N} \rightarrow \mathbb{C}$ such that $\lim_{n \rightarrow \infty} v_f(w_n) = +\infty$ and the w_n are not in the image of $\mathbb{R} \cup \{\infty\}$ with respect to F . We assume, as we may, that the w_n all lie in a component of $\hat{\mathbb{C}} - F(\mathbb{R} \cup \{\infty\})$. But by Theorem 4.1, Ch. VI, the value of v_f is the same at each point of such a component and is finite. We obtain a contradiction. It is to be noticed that implicit in the argument is the precise information that we have on the nature of $F(\mathbb{R} \cup \{\infty\})$. The valence of f is bounded.

We proceed by studying φ near a_k . To that end, let r be a positive number so small that $F(z) \neq F(a_k)$ when

$$0 < |z - a_k| < r, \quad \text{Im } z \geq 0, \quad (4.1)$$

and that $\text{sg}[F(x) - F(a_k)]$ be independent of x in $]a_k, a_k + r[$ (resp. $]a_k - r, a_k[$). The set of z specified by (4.1) will be denoted by Ω . Let l denote a continuous logarithm of $[F - F(a_k)]|_{\Omega}$. The imaginary part of l is constant on $]a_k, a_k + r[$ (resp. on $]a_k - r, a_k[$). Let α_k denote the difference of the common value of $\text{Im } l$ on $]a_k - r, a_k[$ and the common value of $\text{Im } l$ on $]a_k, a_k + r[$. Let λ denote a continuous logarithm of $z - a_k$ with domain Ω . Then $l - (\alpha_k/\pi)\lambda$ is analytic in $\text{int } \Omega$ and its imaginary part takes a constant value at the points of $\Omega \cap \mathbb{R}$. Hence $l - (\alpha_k/\pi)\lambda$ is the restriction to its domain of a function g analytic on $\Delta(a_k; r) - \{a_k\}$ which satisfies $g(\bar{z}) = \overline{g(z)} + ic$, c real.

Suppose that $\alpha_k \neq 0$. We have

$$|f(z) - F(a_k)|^{\pi/\alpha_k} = |(z - a_k)e^{\pi g(z)/\alpha_k}|. \dagger \quad (4.2)$$

Hence $(z - a_k)e^{\pi g(z)/\alpha_k}$ possesses a limit at a_k . We are led to the basic conclusion that g has a *removable singularity* at a_k . It follows on considering $l - (\alpha_k/\pi)\lambda$ that

$$\frac{f'(z)}{f(z) - F(a_k)} = \frac{\alpha_k}{\pi} \frac{1}{z - a_k} + g'(z),$$

$z \in \text{int } \Omega$, and hence we conclude that

$$\frac{f''(z)}{f'(z)} = \left(\frac{\alpha_k}{\pi} - 1 \right) \frac{1}{z - a_k} + h(z), \quad (4.3)$$

$z \in \text{int } \Omega$, where h is analytic at a_k . The role of the nonvanishing of α_k in reaching this conclusion is clear from the calculation involved.

Suppose that $\alpha_k = 0$. In this case we infer that l is the restriction to Ω of a function analytic on $\Delta(a_k; r) - \{a_k\}$ and this function must have a removable singularity at a_k as was the case for g . But then the continuity of F at a_k would

$\dagger$ The notation on the left is in the sense of real analysis: $A^B = \exp(B \log A)$, A positive, B real.

be violated. The case $\alpha_k = 0$ is excluded. We see now from (4.2) that $\alpha_k > 0$. It suffices to consider z near a_k .

We see that we may introduce similarly α_∞ (= the change of a continuous argument of $F(re^{i\theta}) - F(\infty)$ with domain $[0, \pi]$ where r is large and positive). Paraphrasing the above argument we obtain the conclusion that

$$\frac{f''(z)}{f'(z)} = \left(\frac{\alpha_\infty}{\pi} - 1 \right) \frac{1}{z} + h_\infty(z),$$

z large, where h_∞ is analytic at ∞ and takes the value zero there, and $n(\infty; h_\infty) \geq 2$. We conclude from the finiteness of $F(\infty)$ that $\alpha_\infty < 0$.

We now see from the information obtained concerning the behavior of f''/f' near ∞ and the a_k that φ introduced above satisfies

$$\varphi(z) = \sum_1^n \frac{\pi^{-1}\alpha_k - 1}{z - a_k}, \quad (4.4)$$

and further that

$$\sum_1^n \left(\frac{\alpha_k}{\pi} - 1 \right) = \frac{\alpha_\infty}{\pi} - 1. \quad (4.5)$$

Thus we see that a necessary condition that F be an admitted function is that *the logarithmic derivative of the derivative of its restriction to $\{\operatorname{Im} z > 0\}$ have a representation of the form*

$$\frac{f''(z)}{f'(z)} = \sum_1^n \frac{\pi^{-1}\beta_k - 1}{z - a_k}, \quad (4.6)$$

where the β_k are positive and are subject to the condition

$$\sum_1^n \beta_k < (n-1)\pi. \quad (4.7)$$

Let us now turn to the converse question. We shall show that *a nonconstant function f analytic on $\{\operatorname{Im} z > 0\}$ which satisfies the condition (4.6) is the restriction to $\{\operatorname{Im} z > 0\}$ of an admitted F .*

To establish this it will be convenient to introduce the simply connected region A containing $\{\operatorname{Im} z \geq 0\} - \{a_k: 1 \leq k \leq n\}$ defined as

$$\mathbf{C} - \bigcup_{1 \leq k \leq n} \{a_k - it: t \in \mathbf{R}^+\}.$$

We introduce g , a primitive of

$$\sum_1^n \frac{\pi^{-1}\beta_k - 1}{z - a_k},$$

with domain A , and let h denote a primitive of $\exp \circ g$. Since $f'(z) = ce^{g(z)}$, $c \in \mathbb{C} - \{0\}$, it follows that $f(z) = c h(z) + d$, $d \in \mathbb{C}$. We note that

$$\operatorname{Im} g \mid]a_k, a_{k+1}[$$

is constant, $k = 1, \dots, n-1$, and that the same holds for the restriction of $\operatorname{Im} g$ to $\{x < a_1\}$ and the restriction of $\operatorname{Im} g$ to $\{a_n < x\}$. It follows that the image of each of these intervals with respect to h and hence with respect to $ch + d$ lies on a line. Our assertion will follow once we show that $\lim_{\infty} h$ and $\lim_{a_k} h$ exist and are finite. Let us consider the behavior of h near a_k . Let λ denote an analytic logarithm of $z - a_k$ with domain A . Then

$$g(z) = (\pi^{-1}\beta_k - 1)\lambda(z) + \varphi(z)$$

where φ is analytic at a_k . On introducing the Taylor expansion $\sum_0^{\infty} A_j(z - a_k)^j$ of $\exp \circ \varphi$ about a_k , we see that

$$h(z) = \left[\sum_0^{\infty} \frac{A_j}{\pi^{-1}\beta_k + j} (z - a_k)^j \right] \exp[\pi^{-1}\beta_k \lambda(z)] + B, \quad (4.8)$$

$B \in \mathbb{C}$, when z is near a_k . Hence the stated limit property of h at a_k holds. The behavior of h at ∞ is similarly treated. It is to be noted that if λ now denotes an analytic logarithm of $z - a_1$ with domain A , then $g(z) = \alpha \lambda(z) + \varphi(z)$, for z large and satisfying $\operatorname{Im} z > 0$, where φ is analytic at ∞ and $\alpha < -1$. This follows on considering the right-hand side of (4.6) near ∞ and taking (4.7) into account. The remainder of the argument parallels that just given for the behavior at a_k . We introduce the Laurent expansion of $\exp \circ \varphi$ in $\{|z - a_1| > r\}$, r large, and verify that a counterpart of (4.8) now holds.

EXERCISES

4.1. Study with the aid of the method of Schwarz–Christoffel treated above the continuous univalent maps of $\{\operatorname{Im} z \geq 0\} \cup \{\infty\}$ onto a rectangle which are analytic in $\{\operatorname{Im} z > 0\}$.

4.2. Let $\operatorname{Im} a > 0$. Let $T = \{\lambda + \mu a; \lambda, \mu \geq 0, \lambda + \mu \leq 1\}$. Show that there is a unique Schwarz–Christoffel function F mapping $\{\operatorname{Im} z \geq 0\} \cup \{\infty\}$ univalently onto T and satisfying $F(0) = 0$, $F(1) = 1$, $F(\infty) = a$. Determine F .

4.3. Extend the study of the Schwarz–Christoffel functions to the class of continuous maps F of $\{\operatorname{Im} z \geq 0\} \cup \{\infty\}$ into $\hat{\mathbb{C}}$ which satisfy the condition that there exists $(a_k)_{1 \leq k \leq n}$, an increasing sequence of real numbers, such that (1) F is the restriction to its domain of a function meromorphic at each point of

$$\{\operatorname{Im} z \geq 0\} - \{a_k; 1 \leq k \leq n\}$$

which is analytic at each real point not an a_k and has a nonzero derivative at each real point (not an a_k), and (2) $F([a_k, a_{k+1}])$ is contained in a line, $k = 1, \dots, n-1$, and the same is true for $F(\{x < a_1\})$ and $F(\{x > a_n\})$. Distinguish the special cases:

F analytic in $\{\operatorname{Im} z > 0\}$, F locally univalent in $\{\operatorname{Im} z > 0\}$, F both analytic and locally univalent in $\{\operatorname{Im} z > 0\}$.

5. REGIONS WITH SIMPLE POLYGONAL FRONTIERS

Let $(t_k)_{0 \leq k \leq n}$ be increasing and satisfy $t_0 = 0$, $t_n = 1$. Let $a_k, b_k \in \mathbb{C}$, $k = 0, \dots, n-1$, and satisfy $a_k t_{k+1} + b_k = a_{k+1} t_{k+1} + b_{k+1}$, $k = 0, \dots, n-2$, as well as $b_0 = a_{n-1} + b_{n-1}$. Let Γ denote the closed curve in $\mathbb{C}$ satisfying $\Gamma(t) = a_k t + b_k$, $t_k \leq t \leq t_{k+1}$, $k = 0, \dots, n-1$. By a *polygonal closed curve* is meant such a Γ (image as well as map). The $\Gamma(t_k)$ are termed the *vertices* of Γ . Γ is termed *simple* provided that $\Gamma|_{[0, 1[}$ is univalent. Let A be a region in $\mathbb{C}$ whose frontier in the sense of $\hat{\mathbb{C}}$ is Γ assumed simple. Our goal is to show the following theorem.

Theorem 5.1: *A is holomorphically simply connected. A univalent function f analytic on $\{\operatorname{Im} z > 0\}$ having A as its image is the restriction of a univalent Schwarz–Christoffel function F . If $F(\infty)$ is a vertex of Γ , the α_k and α_∞ associated with the preimages with respect to F of the vertices satisfy*

$$0 < \alpha_k < 2\pi, \quad k = 1, \dots, n-1,$$

as well as

$$-2\pi < \alpha_\infty < 0$$

and

$$\sum_{k=1}^{n-1} \alpha_k = (n-1)\pi + \alpha_\infty.$$

PROOF: The first assertion follows from the fact that $\hat{\mathbb{C}} - A$ is connected. This is a general property enjoyed by regions in $\mathbb{C}$ whose frontiers (in the sense of $\hat{\mathbb{C}}$) are connected. For the sake of completeness we include a proof. Let Ω be such a region in $\mathbb{C}$ with frontier β . We note that $\hat{\mathbb{C}} - \Omega$ is the union of β and the components of $\hat{\mathbb{C}} - \beta$ other than Ω (which is itself a component of $\hat{\mathbb{C}} - \beta$). Now the components of $\hat{\mathbb{C}} - \beta$ have the property that their frontiers lie in β . Hence $\hat{\mathbb{C}} - \Omega$ is the union of β and the closures of the components of $\hat{\mathbb{C}} - \beta$ other than Ω . We shall now use the following two properties of connected sets in a topological space: Let X be a topological space, let $Y, Y_\lambda (\subset X)$ be connected, $\lambda \in \Lambda$. Then (1) $\bar{Y}$ is connected (Ex. 3.1, Ch. III), (2) $\bigcup_{\lambda \in \Lambda} Y_\lambda$ is connected when $\bigcap_{\lambda \in \Lambda} Y_\lambda \neq \emptyset$ (Theorem 3.2, Ch. III). It follows, since the closure of a component of $\hat{\mathbb{C}} - \beta$ is connected and has a point in common with β , that its union with β is connected. Hence the union of β and the union of the

unions of β and the closures of the components of $\hat{C} - \beta$ taken with respect to the components of $\hat{C} - \beta$ other than Ω is connected. It is equal to $\hat{C} - \Omega$.

We introduce B , the component of $\hat{C} - \Gamma$ containing ∞ . We show that $\text{fr } B = \Gamma$. At all events, we know that $\text{fr } B \neq \emptyset$. Suppose that $z \in \Gamma$. Thanks to the simple polygonal nature of Γ , when r is sufficiently small, $\Delta(z; r) - \Gamma$ has exactly two components the frontier of each of which contains $\Delta(z; r) \cap \Gamma$. The set of $z \in \Gamma$ for which one of these components is contained in B (for all small r) is not empty as we see from the fact that $\text{fr } B$ is not empty and is contained in Γ . We see that the set of $z \in \Gamma$ having the property stated in the preceding sentence is *open* in the sense of the relative topology of Γ . This is concluded by examining the set $\Delta(z; r) - \Gamma$ directly when r is small. The details are readily supplied. That the set in question is *closed* is immediate. Hence the set is Γ since Γ is connected, and we conclude that $\text{fr } B = \Gamma$.

We turn to the remaining assertions of Theorem 5.1. We are, of course, guaranteed by the Riemann mapping theorem that there exists a univalent function g analytic on $\Delta(0; 1)$ having image A . Let h denote the inverse of g . We proceed by considering the behavior of h near a point of Γ . From the fact that $\lim_z |h| = 1$, $z \in \Gamma$ (which follows from the fact that h is a univalent function analytic on A and having image $\Delta(0; 1)$), we see at once that $\lim_z \log |h| = 0$, $z \in \Gamma$. We conclude that if ζ is *not* a vertex of Γ , then there exists by virtue of the Schwarz reflexion principle and the remarks of the preceding paragraph a function H analytic on a disk $\Delta(\zeta; r)$ such that $H(z) = h(z)$, $z \in A \cap \Delta(\zeta; r)$. Hence $\lim_{\zeta} h$ exists. Also $h[A \cap \Delta(\zeta; r)]$ is the intersection of $\Delta(0; 1)$ and a neighborhood of $\lim_{\zeta} h$. In addition, $H'(\zeta) \neq 0$ for otherwise $|h|$ would take values greater than 1 at points near ζ .

Suppose that ζ is a vertex of Γ . We take r so small that the component C of $\Delta(\zeta; r) - \Gamma$ which lies in A is a set of the form

$$\{\zeta + e^z: \text{Re } z < \log r, \theta_1 < \text{Im } z < \theta_2\}$$

where $\theta_1 < \theta_2 < \theta_1 + 2\pi$. We introduce

$$\varphi(z) = \zeta + \exp \left[\frac{\theta_2 - \theta_1}{\pi} \text{Log } z + i \frac{\theta_1 + \theta_2}{2} \right]$$

with domain $\{\text{Re } z > 0\} \cap \Delta(0; \rho)$ where $\log \rho = (\pi \log r)/(\theta_2 - \theta_1)$. It is readily verified that φ is a univalent function analytic on its domain and having image C . Further φ is the restriction of a univalent function Φ continuous on the closure of the domain of φ . We note that $\Phi(0) = \zeta$ and the image of $\{it: -\rho < t < \rho\}$ with respect to Φ is $\Delta(\zeta; r) \cap \Gamma$. On now considering $h \circ \varphi$ near 0, we see as above that h possesses a limit ω at ζ , that the image of $h \circ \varphi$ is the intersection of $\Delta(0; 1)$ and a neighborhood of ω , and that the same holds true for the image of $h| [A \cap \Delta(\zeta; r)]$. We conclude that h is the

restriction to A of a continuous map H of $\bar{A}$ onto $\overline{\Delta(0; 1)}$. Now H is univalent. Otherwise by the remarked nature of $h[A \cap \Delta(\zeta; r)]$ we would be forced to conclude that h was not univalent. The inverse of H is a univalent continuous map of $\overline{\Delta(0; 1)}$ onto $\bar{A}$ whose restriction to $\Delta(0; 1)$ is g . Further g is the restriction of a function G which is analytic and has a nonzero derivative at each point of $\overline{\Delta(0; 1)}$ less the image with respect to H of the vertices of Γ .

The function

$$g\left(\frac{z-i}{z+i}\right)$$

with domain $\{\operatorname{Im} z > 0\}$ is an f of the theorem. The others are obtained by composition with automorphisms of $\{\operatorname{Im} z > 0\}$ analytic on $\{\operatorname{Im} z > 0\}$. The Schwarz–Christoffel character of the f follows.

There remains only to verify the inequalities stated for α_∞ and the α_k , the asserted equality following from (4.5). We note that $F(z)$ admits a representation of the form $\Phi[\Psi(z)]$ near the preimage a with respect to F of a vertex ζ where Ψ is analytic at a , $\Psi(a) = 0$, $n(a; \Psi) = 1$, $\operatorname{Re} \Psi(x) = 0$ for x real near a . Consulting the definition of the α_k and α_∞ , we conclude that $\alpha_k = \theta_2 - \theta_1$ (using the notation employed in the definition of φ with $\zeta = F(a_k)$) and similarly, understanding the corresponding conventions, that $-\alpha_\infty = \theta_2 - \theta_2$. The assertion follows.

EXERCISE

5.1. Analyze in greater detail the geometric aspects of the above exposition. In particular, the nature of the component C of $\Delta(\zeta; r) \cap \Gamma$ should be examined. Supply the argument relating α_k (resp. α_∞) and $\theta_2 - \theta_1$. Here one may proceed by studying changes of continuous arguments.